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Is Cash the King of (Promised) Incentives for All Subpopulations?
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Abstract

Common wisdom in survey methodology is that cash is the most persuasive form of incentive payment. Since the establishment of this guideline, however, the world has become more digital. Restaurants and retailers have started to move away from accepting cash payments, a trend accelerated by the pandemic. Gift cards, especially those issued by online retailers, are often digital. Smart phones allow people to leave their wallets at home and instead use their phones to pay-- either digitally with an app or with a credit card tucked into a card holder stuck to the phone. Given these shifts in how people pay for things, is cash still enticing to all subpopulations?

Data collected from a large nonprobability survey asked about people's use of cash and their preferred form of incentive payment for \$5 and \$40. Incentive types offered in the vignette were: cash, digital gift cards, physical gift cards, check, and electronic bank transfer. A small number of people preferred cash incentives for either amount (5%), with most opting for electronic bank transfers and digital gift cards. The small subpopulation preferring cash was mostly young, low-income, and without a college degree. It seems that cash is no longer king but digital cash (i.e., digital gift cards and bank transfers) now reigns.

Keywords: incentives, incentive preference, digital incentives

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Overview

The common wisdom among survey methodologists is that cash incentives are more effective at gaining response than alternative monetary incentives (e.g., gift cards) and non-monetary incentives (e.g., Gajic et al. 2012, Birnholtz et al. 2004, Singer and Ye 2013). This research has led to the mantra that “cash is king” in the incentive world. While cash is more motivating than a pen or magnet, in a post-Covid world, how does it compare to alternative monetary incentives? It may be time to update the mantra.

Changes to consumers’ relationship with cash were accelerated with the ubiquity of contactless payment methods that arose during Covid (Dickler 2021). Beyond handing over a credit card, using one’s smartphone as a digital wallet can eliminate the need to carry a physical wallet, and therefore perhaps the ability to carry cash. So, is it now inconvenient to respondents to offer cash if their habits have changed such that they don’t carry it or use it?

There is evidence that the enthusiasm to abandon cash is likely not evenly distributed across the population, and may vary by income, race, and age. Studies conducted by Pew Research in 2015, 2018, and 2022 (Perrin 2018; Faverio 2022) reported that low-income households are more likely to use cash, compared to mid- and high-income households. Black consumers are more likely to use cash than White consumers, and people ages fifty and older are more likely to pay with cash, compared to people younger than fifty. Given this, receptiveness to a cash incentive may vary by demographics, leading to different preferences for different subpopulations.

This research collects information on respondents’ relationship with cash and asks vignette questions about how they would like to receive \$5 and \$40 promised incentives for completing a web survey. The results will show respondents’ incentive preferences and characterize those who still prefer a cash incentive.

Research Questions

1. Are people carrying cash? Of those carrying cash, what are their characteristics?
2. Are they using cash? Of those using cash, what are their characteristics?
3. What type of incentive payment is preferred for \$5 vs. \$40?
4. What are the characteristics of those who prefer cash incentives?
5. Overall, is cash, as a promised incentive, still enticing for all subpopulations?

Data

The experiment was hosted on a web survey of Census Bureau Research Panel members. These 30,000 panel members were respondents to Phase 4.2 of the Census Household Pulse survey (HPS), administered from July to September 2024, who agreed to be contacted about future research opportunities. The web survey was open from 3-15

December 2024 and available in English only. The survey asked about demographics, household composition, carrying and spending cash, preferences for incentive payment method, costs of utilities, school lunches, and income from property rental (in that order). It took approximately 15 minutes. Two reminders were sent, beyond the initial email invitation, resulting in approximately 10,000 responses and 500 bounce backs (assumed eligible), for an AAPOR RR2 of 33%. There were about 9,200 cases still active in the survey for the questions about incentive preference and having and using cash. No incentive was offered for completing the survey.

Applicable Survey Questions

Respondents were randomized into two groups to determine which incentive vignette they were shown first. For illustration, a screenshot of the \$40 incentive vignette question is shown in Figure 1 (the other group would see the same question but with “\$5” instead). The respondent could choose to receive cash, digital gift card, physical gift card, check, or electronic bank transfer, or decline payment. Note that incentive options include how the incentive would be delivered, in case that is a factor in the respondent’s decision. The survey length of an hour is included in the vignette to encourage respondents to take a payment for their time (and to avoid confusion about receiving an incentive for the shorter survey they were currently answering). After answering about either \$40 or \$5 (depending on their group), they were asked if their answer would change for the other incentive amount. If yes, they were asked their preference for the new amount.

Figure 1: The series of incentive preference questions, for those shown \$40 first

Imagine that you just completed a survey online that took an hour. The last question of the survey asks how you would like to receive your \$40 payment. Which method would you choose?

- ☐ Cash, mailed to your home
- ☐ Digital gift card, emailed
- ☐ Physical gift card, mailed to your home
- ☐ Check, mailed to your home
- ☐ Electronic transfer to your bank account
- ☐ I wouldn't take the payment

Would your answer change if the payment was \$5?

- ☐ Yes
- ☐ No

[If yes, the answer would change]

If you completed a survey online that took an hour, how would you want to receive the \$5 payment?

- ☐ Cash, mailed to your home
- ☐ Digital gift card, emailed
- ☐ Physical gift card, mailed to your home
- ☐ Check, mailed to your home
- ☐ Electronic transfer to your bank account
- ☐ I wouldn't take the payment

Additional survey questions relevant to the analysis were:

- If a friend owes you \$40, how do you prefer that they pay you back? {cash, payment app (e.g., Venmo, Zelle, Paypal), check, gift card}
- When you receive a check in the mail, how do you usually cash or deposit the check? {ATM, In-person at the bank, Bank's app on my phone, Payment app (e.g., Venmo), Store that cashes checks, Sign it over to someone else}
- Do you ever use an app to do your banking? { Yes, No}
- Do you ever use an app to pay people? { Yes, No}
- How often do you have cash? {Always, Most of the time, Sometimes, Rarely, Never} Question includes a checkbox for "I only have cash for emergencies" under the response options that can be selected in addition to a response.
- How often do you use cash to pay for things? {Always, Most of the time, Sometimes, Rarely, Never}

As noted, respondents were asked how often they have cash, using a Likert scale from Always to Never, but could also designate they have cash "only for emergencies" in a checkbox below the Likert scale responses. Among those who checked the box, "Rarely" had the highest frequency of the five categories. Therefore, the *thirty* people who checked this box but did not also select a Likert scale response were recoded as "Rarely" so that they did not drop out of the analysis.

The analysis controlled for the following demographic variables: age (19-35, 36-54, 55-70, 71-96), education (less than a college degree; college degree; graduate degree), sex (flag for male), race (White alone, Hispanic or Latino alone, Black or African American alone, other), household income (less than \$35,000, \$35,000-\$99,999, \$100,000 or more), flag for children under 18 years old in the household, flag for currently working, and flag for using a computer daily.

Methods

For the research questions about carrying and using cash, bivariate distributions are presented followed by multivariate relationships using logistic regression. Given the amount of missing data for household income, models were run first without income. For the final model, predicting who prefers cash incentives, the logistic regression drops several variables that are not significant, like children in the household and frequency of carrying cash. This refined model is used to generate propensity scores to more clearly characterize who is more likely to prefer a cash incentive.

Results

RQ1: Are people carrying cash? Of those carrying cash, what are their characteristics?

Table 1 shows a fairly even distribution of responses from Always to Rarely, indicating that most people (52%) often carry cash but very few people Never have cash on hand. Among those who checked the box that they only carry cash for emergencies, most report Rarely having cash (44%).

Table 1: Responses to questions about carrying and using cash

	Have Cash n=9,200	“Only for Emergency” n=550	Use Cash n=9,200
Always	30%	18%	1%
Most of the time	22%	17%	5%
Sometimes	21%	13%	35%
Rarely	25%	44%	55%
Never	2%	8%	5%

Note: Table 1 includes cases that only selected “only for emergency”, recoded as Rarely.

A logistic regression predicting carrying cash (see Table 2) finds that the older a person is (using broad categories), the more likely they are to carry cash. Males, people with more education, and people who are white (alone) are more likely than their counterparts to carry cash. Those less likely to carry cash are people currently working and people with children under eighteen years old in the household. When income is added to the model, there are only small changes to the odds ratios, and results indicate that those with higher household incomes are more likely to carry cash.

Table 2: Logistic regressions predicting carrying cash “Always” or “Most of the time”

	Model 1		Model 2 - with Income	
	n=9,100		n=8,700	
	Odds ratio		Odds ratio	
Age				
19 - 35	(ref)		(ref)	
36 - 54	1.33	***	1.32	***
55 - 70	3.00	***	2.98	***
71 - 96	5.12	***	5.11	***
Working	0.79	***	0.75	***
Use computer daily	1.03		1.01	
Education				
No college degree	(ref)		(ref)	
College degree	1.36	***	1.24	***
Grad degree	1.42	***	1.27	***
Male	1.76	***	1.70	***
Child <18 in HH	0.78	***	0.76	***
Race				
White alone	(ref)		(ref)	
Hispanic alone	0.64	**	0.65	**
Black alone	0.44	***	0.49	***
Other	0.81	**	0.82	**
HH Income				
<\$35K			(ref)	
\$35k-\$99,999			1.50	***
\$100k+			1.72	***

*** p<0.001; ** p<0.01

RQ2: Are they using cash? Of those using cash, what are their characteristics?

Respondents were asked how often they use cash, using the same Likert scale as carrying cash. The responses are more skewed, with majority of people reporting that they Rarely (55%) or Sometimes (35%) use cash (see Table 1). Comparing those who carry cash to those who use cash, it is evident that people are carrying cash but not using it often. Instead of high cash use among those who often carry cash, this incidence is only 5% (Table 3). Most people who carry cash Always or Most of the time report using it Sometimes, Rarely, or Never (48%).

Table 3: Cross-tabulation of carrying cash and using cash

		Use Cash	
		Not often	Always/ Most
Have Cash		n=8,700	n=500
Not often	n=4,400	47.7%	0.5%
Always/Most	n=4,800	47.9%	4.9%

$\chi^2(1)=312$; $p<0.001$

A logistic regression predicting use of cash (see Table 4) shows that, like carrying cash, older people are more likely to use cash, compared to younger people (19 -35 years old). However, working and having children in the household are not significant correlates. In addition, the relationship between cash use and education, race, and income is opposite of the relationship between carrying cash and these variables (i.e., White people, more educated people, and people with higher household incomes are less likely to use cash).

Table 4: Logistic regressions predicting using cash “Always” or “Most of the time”

	Model 3		Model 4- with Income	
	n=9,100		n=8,700	
	Odds ratio		Odds ratio	
Age				
19 - 35	(ref)		(ref)	
36 - 54	2.45	***	2.76	***
55 - 70	3.31	***	3.97	***
71 - 96	3.46	***	3.90	***
Working	0.95		1.07	
Use computer daily	0.47	***	0.53	***
Education				
No college degree	(ref)		(ref)	
College degree	0.48	***	0.56	***
Grad degree	0.37	***	0.51	***
Male	1.25	*	1.39	***
Child <18 in HH	0.82		0.97	
Race				
White alone	(ref)		(ref)	
Hispanic	1.64	*	1.39	
Black alone	2.60	***	2.34	***
Other	1.11		1.00	
HH Income				
<\$35K			(ref)	
\$35k-\$99,999			0.68	***
\$100k+			0.34	***

*** p<0.001; **p<0.01; *p<0.05

RQ3: What type of incentive payment is preferred for \$5 vs. \$40?

Respondents were randomly assigned a question about a \$5 or \$40 incentive and then asked if their answer would change for the other amount. The assignment was intended to balance confirmation bias, in case the amount asked first influenced the preference for the other amount. Of those asked about \$40 first, 17% changed their answer for \$5, showing that respondents were consistent about their incentive preference. Table 5 shows small but significant differences in the preferences for the \$5 incentive by what was asked first. For example, people asked about the \$40 incentive first were less likely to want a digital gift card for \$5 and more likely to want a check.

Table 5: Incentive preference for five-dollar incentive

	Asked \$5 First n=4,600	Asked \$40 First n=4,600	All cases n=9,200
Cash, mailed	7%	6%	6%
Digital gift card, emailed	32%	27%	29%
Physical gift card, mailed	6%	6%	6%
Check, mailed	16%	22%	19%
E- transfer to bank	29%	28%	29%
Wouldn't take payment	10%	11%	10%
	$\chi^2(5)=50; p<0.001$		

Similarly, Table 6 shows small but significant differences in preference for the \$40 incentive depending on what was asked first. More people who were asked about the \$5 incentive first chose a digital gift card for the \$40 (this corresponds to the difference reported above, for \$5). Of those asked about \$5 first, 14% changed their answer for \$40, again showing consistency in incentive preference.

Table 6: Incentive preference for forty-dollar incentive

	Asked \$40 First n=4,600	Asked \$5 First n=4,600	All cases n=9,200
Cash, mailed	4%	4%	4%
Digital gift card, emailed	26%	32%	29%
Physical gift card, mailed	7%	7%	7%
Check, mailed	25%	21%	23%
E- transfer to bank	34%	32%	33%
Wouldn't take payment	4%	3%	4%
	$\chi^2(5)=54; p<0.001$		

Table 7 combines all responses to the \$5 or \$40 incentive questions, regardless of whether asked first or second, and provides an average for the incentive type, across the two amounts. This table clearly shows that digital gift cards and electronic bank transfers are the most preferred options, regardless of incentive amount. Digital gift cards were preferred by 29% of respondents for both amounts, and bank transfers were preferred by 29% (\$5) or 33% (\$40) of respondents.

Cash is the least popular incentive, with 6% (\$5) or 4% (\$40) of cases preferring this, depending on the incentive amount. Physical gift cards are also not very popular (6% or 7% of cases). Both were designated in the vignette as delivered in the mail, which may seem like a driving factor except that 19% (\$5) or 23% (\$40) preferred a (mailed) check, which approaches the popularity of the digital gift cards.

Table 7: All responses for five- and forty- dollar incentives, regardless of whether asked first or second

	\$5 n=9,200	\$40 n=9,200	Average
Cash, mailed	6%	4%	5%
Digital gift card, emailed	29%	29%	29%
Physical gift card, mailed	6%	7%	6%
Check, mailed	19%	23%	21%
E- transfer to bank	29%	33%	31%
Wouldn't take payment	10%	4%	7%

It is notable that, on average, 7% of respondents said they would not take an incentive, with more people providing this response for the \$5 incentive. Table 8 shows that when people chose different responses for the two incentive amounts, they most often reported that they would not take a payment for the \$5 incentive, compared to choosing a different incentive type for the smaller payment.

The survey that collected the data for this experiment did not include an incentive for response. Even though the vignette noted an hour-long survey (compared to fifteen minutes to complete the active survey), respondents willing to respond to this survey may be more likely not to take an incentive than the general population.

Table 8: Comparison of incentive preference for forty-dollars to preference for five-dollars

Preference when \$5	Preference when \$40					
	Cash	Digital gift card	Physical gift card	Check	E-transfer to bank	Wouldn't take payment
	n=350	n=2,700	n=600	n=2,100	n=3,100	n=350
Cash	91%	2%	5%	5%	2%	
Digital gift card	1%	90%	3%	5%	5%	
Physical gift card	1%		80%	1%	1%	
Check	1%		1%	79%	2%	
E-transfer to bank	1%	1%			86%	
Wouldn't take payment	5%	7%	11%	10%	4%	100%

RQ4: What are the characteristics of those who prefer cash incentives?

Table 9 shows a logistic regression to describe the small percentage of cases who preferred cash, for either incentive amount. Although both carrying and using cash were included in the model, only using cash is a significant predictor. In addition, this model includes a variable indicating that if a friend owed the respondent \$40, they would prefer to be paid back in cash, rather than check, payment app, or gift card. Use of apps for payment or banking is also a significant, but negative, correlate of choosing a cash incentive.

Compared to carrying or using cash, the correlation between age and choosing a cash incentive is opposite, with the youngest age group now being most likely to choose cash. People with a college degree are significantly less likely to choose cash than those without. Those with the lowest household income (less than \$35,000) are more likely to use cash than people with higher incomes.

Table 9: Logistic regression predicting preference for cash incentive for either five- or forty-dollars

Chose Cash		
n=8,700		
odds ratio		
Use Cash		
Sometimes-Never	(ref)	
Always/Most of time	1.72	***
Paid Cash if owed money	2.44	***
Use bank/cash apps	0.64	***
Age		
19 – 35 years	(ref)	
36 – 54 years	0.63	***
55 – 70 years	0.55	***
71 – 96 years	0.53	***
Education		
No college degree	(ref)	
College degree	0.67	***
Grad degree	0.85	
Male	1.04	
HH Income		
<\$35K	(ref)	
\$35k-\$99,999	0.72	**
\$100k+	0.67	**

*** p<0.001; **p<0.01

Propensity scores are helpful for combining the different significant attributes and characterizing the people who prefer cash incentives. Highlighting propensities of 0.10 or above, it appears that low household income (less than \$35,000) is driving this preference, as it is the common factor for almost all propensities of 0.10 or higher. The low-income group has a small sample size for many of the age and degree combinations which may mean that a few influential cases are driving this result for some groups.

Table 10: Propensity scores for combinations of different demographic characteristics, based on model predicting preference for a cash incentive

		n	Mean Propensity
Age			
19 - 35			
No college degree			
	<\$35K	80	0.14
	\$35k-\$99,999	150	0.10
	\$100k+	70	0.08
College degree			
	<\$35K	40	0.07
	\$35k-\$99,999	200	0.06
	\$100k+	300	0.05
Grad degree			
	<\$35K	20	0.10
	\$35k-\$99,999	100	0.07
	\$100k+	250	0.06
36 - 54			
No college degree			
	<\$35K	200	0.13
	\$35k-\$99,999	350	0.08
	\$100k+	250	0.06
College degree			
	<\$35K	60	0.08
	\$35k-\$99,999	300	0.05
	\$100k+	650	0.04
Grad degree			
	<\$35K	40	0.09
	\$35k-\$99,999	250	0.06
	\$100k+	1,000	0.04
55-70			
No college degree			
	<\$35K	300	0.13
	\$35k-\$99,999	500	0.09
	\$100k+	300	0.07
College degree			
	<\$35K	100	0.08
	\$35k-\$99,999	300	0.06
	\$100k+	400	0.04
Grad degree			
	<\$35K	60	0.10
	\$35k-\$99,999	250	0.07
	\$100k+	600	0.05
71-96			
No college degree			
	<\$35K	150	0.13
	\$35k-\$99,999	250	0.09

	\$100k+	90	0.08
College degree	<\$35K	70	0.09
	\$35k-\$99,999	200	0.06
	\$100k+	150	0.05
Grad degree	<\$35K	50	0.11
	\$35k-\$99,999	250	0.07
	\$100k+	350	0.06

Conclusion: Overall, is cash, as a promised incentive, still enticing for all subpopulations?

The analysis found that although people carry cash, they are not using it often. Also, those who do use it often have different characteristics than those carrying it often.

Promised cash incentives were preferred by only 5% of respondents, with digital gift cards and electronic bank transfers being most popular. Of those who wanted to receive a cash incentive, they were more likely to be young (19- 35 years old), not have a college degree, be low-income (<\$35,000), and use cash often. Analysis of the propensity scores for different demographic groups revealed that having a low income was the most consistent factor.

The incentive options provided to respondents in the vignette were all forms of monetary payment, with only one option being physical dollar bills (i.e., cash). Therefore, this study provides clarification as to respondent preferences for receiving monetary incentives. Cash, in the form of dollars, is not king but digital money is appealing (i.e., digital gift cards and bank transfers). It seems that the mantra should be updated to, “Digital cash is king”, at least for promised incentives.

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